

Experiment No. 1

Experiment Name

Real-World Data Collection, Cleaning and Preprocessing using Python

Aim

To collect a real-world dataset from publicly available sources and perform data preprocessing by handling missing values, duplicate records, inconsistent formats, categorical variables, and outliers using Python.

Purpose

In real-world analytics projects, data collected from multiple sources is often incomplete, inconsistent, and noisy. Before performing visualization or machine learning, it is essential to preprocess the data to improve quality and reliability.

In this experiment, students will work on the Titanic Survival Dataset or IBM HR Employee Attrition Dataset to understand industry-standard data cleaning techniques such as missing value treatment, duplicate removal, outlier detection, feature encoding, and normalization.

Prerequisites

- Basic knowledge of Python programming
- Familiarity with Pandas and NumPy libraries
- Understanding of CSV file handling
- Jupyter Notebook or Google Colab
- Basic concepts of statistics (mean, median, standard deviation)

Steps

1. Download a real-world dataset such as the Titanic Survival Dataset or IBM HR Employee Attrition Dataset from Kaggle.
2. Import the required libraries (Pandas, NumPy, Matplotlib, and Seaborn).
3. Load the dataset into a Pandas DataFrame and inspect its structure using `head()`, `info()`, and `describe()`.
4. Identify missing values and handle them using appropriate techniques such as mean, median, mode, or row/column removal.
5. Detect and remove duplicate records to ensure data consistency.
6. Standardize inconsistent categorical values and convert them into numerical format using Label Encoding or One-Hot Encoding.

7. Identify outliers in numerical features using Box Plots and the IQR method, and treat them appropriately.
8. Normalize or standardize numerical attributes to bring them to a common scale.
9. Perform feature engineering by creating new attributes (e.g., Family Size, Age Group, or Income Category).
10. Save the cleaned and preprocessed dataset as a new CSV file for further analysis.

Questions

1. Why is data preprocessing considered one of the most important phases in data analytics?
2. Explain different methods of handling missing values with suitable examples.
3. Differentiate between Label Encoding and One-Hot Encoding.
4. What are outliers? How can they affect analytical results?
5. Explain the difference between normalization and standardization.
6. Why should duplicate records be removed before analysis?
7. What is feature engineering? Give two practical examples.
8. Which preprocessing techniques would you apply to the IBM HR Employee Attrition dataset and why?
9. How does poor-quality data affect machine learning model performance?
10. Name any three Python libraries commonly used for data preprocessing.